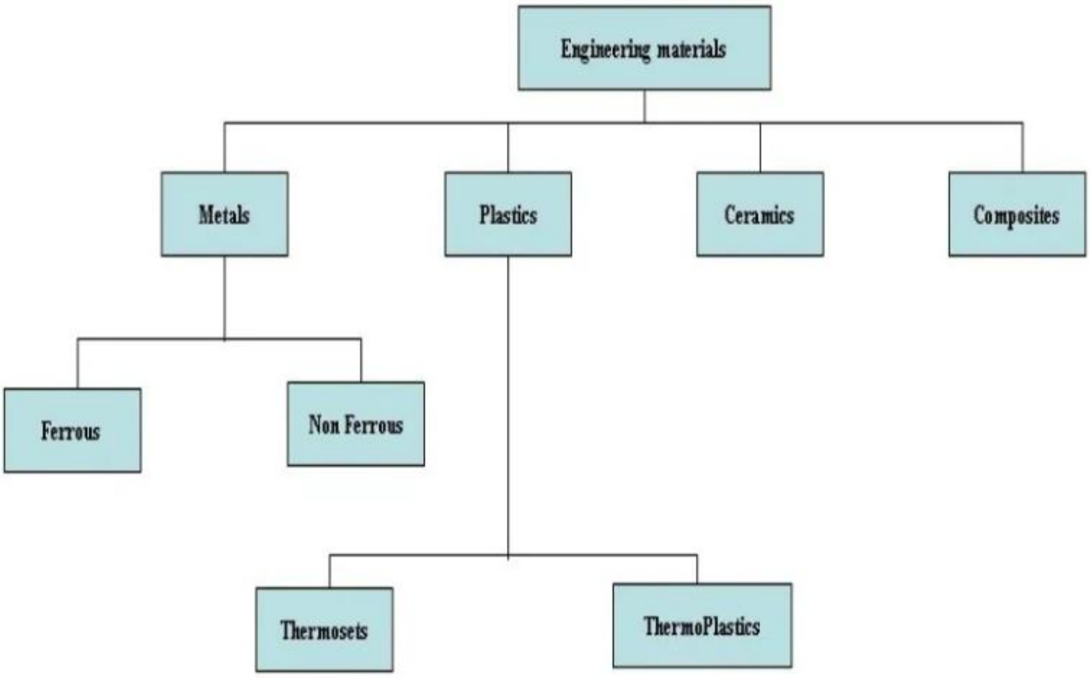
UNIT-1

Engineering Materials and Metal Joining Processes

Engineering materials refers to the group of materials that are used in the construction of manmade structures and components. The primary function of an engineering material is to withstand applied loading without breaking and without exhibiting excessive deflection

Classification of Engineering Materials



Metals

Metals are the most commonly used class of engineering material. Metal alloys are especially common, and they are formed by combining a metal with one or more other metallic and/or non-metallic materials. The combination usually occurs through a process of melting, mixing, and cooling. The goal of alloying is to improve the properties of the base material in some desirable way.

Metal alloy compositions are described in terms of the percentages of the various elements in the alloy, where the percentages are measured by weight.

Ferrous Alloys

Ferrous alloys have iron as the base element. These alloys and include steels and cast irons. Ferrous alloys are the most common metal alloys in use due to the abundance of iron, ease of production, and high versatility of the material. The biggest disadvantage of many ferrous alloys is low corrosion resistance.

Carbon is an important alloying element in all ferrous alloys. In general, higher levels of carbon increase strength and hardness, and decrease ductility and **weldability.**

Carbon Steel

Carbon steels are basically just mixtures of iron and carbon. They may contain small amounts of other elements, but carbon is the primary alloying ingredient. The effect of adding carbon is an increase in strength and hardness.

Most carbon steels are plain carbon steels, of which there are several types.

Low-Carbon Steel

Low-carbon steel has less than about 0.30% carbon. It is characterized by low strength but high ductility. Some strengthening can be achieved through cold working, but it does not respond well to heat treatment. Low-carbon steel is very weldable and is inexpensive to produce. Common uses for low-carbon steel include wire, structural shapes, machine parts, and sheet metal.

Medium-Carbon Steel

Medium-carbon steel contains between about 0.30% to 0.70% carbon. It can be heat treated to increase strength, especially with the higher carbon contents. Medium-carbon steel is frequently used for axles, gears, shafts, and machine parts.

High-Carbon Steel

High-carbon steel contains between about 0.70% to 1.40% carbon. It has high strength but low ductility. Common uses include drills, cutting tools, knives, and springs.

Aluminum Alloys

Pure aluminum is soft and weak, but it can be alloyed to increase strength. Pure aluminum has good corrosion resistance due to an oxide coating that forms over the material and prevents oxidation. Alloying the aluminum tends to reduce its corrosion resistance

Aluminum is a widely used material, particularly in the aerospace industry, due to its light weight and corrosion resistance. Despite the fact that aluminum alloys are generally not as strong as steels, they nevertheless have a good strength-to-weight ratio.

Copper Alloys

Copper alloys are generally characterized as being electrically conductive, having good corrosion resistance, and being relatively easy to form and cast. While they are a useful engineering material, copper alloys are also very attractive and are commonly used in decorative applications.

Copper alloys primarily consist of brasses and bronzes. Zinc is the major alloying ingredient in brass. Tin is a major alloying element in most bronzes. Bronzes may also contain aluminum, nickel, zinc, silicon, and other elements. The bronzes are typically stronger than the brasses while still maintaining good corrosion resistance.

Polymers

Polymers are materials that consist of molecules formed by long chains of repeating units. They may be natural or synthetic. Many useful engineering materials are polymers, such as plastics, rubbers, fibers, adhesives, and coatings. Polymers are classified as thermoplastic polymers, thermosetting polymers (thermosets), and elastomers.

Thermoplastic Polymers

The classification of thermoplastics and thermosets is based on their response to heat. If heat is applied to a thermoplastic, it will soften and melt. Once it is cooled, it will return to solid form. Thermoplastics do not experience any chemical change through repeated heating and cooling (unless the temperature is high enough to break the molecular bonds). They are therefore very well suited to injection molding.

Thermosetting Polymers

Thermosets are typically heated during initial processing, after which they become permanently hard. Thermosets will not melt upon reheating. If the applied heat becomes extreme however, the thermoset will degrade due to breaking of the molecular bonds. Thermosets typically have greater hardness and strength than thermoplastics. They also typically have better dimensional stability than thermoplastics, meaning that they are better at maintaining their original dimensions when subjected to temperature and moisture changes

Elastomers

Elastomers are highly elastic polymers with mechanical properties similar to rubber. Elastomers are commonly used for seals, adhesives, hoses, belts, and other flexible parts. The strength and stiffness of rubber can be increased through a process called vulcanization, which involves adding sulfur and subjecting the material to high temperature and pressure. This process causes cross-links to form between the polymer chains.

Ceramics

Ceramics are solid compounds that may consist of metallic or nonmetallic elements. The primary classifications of ceramics include glasses, cements, clay products, refractories, and abrasives

Ceramics generally have excellent corrosion and wear resistance, high melting temperature, high stiffness, and low electrical and thermal conductivity. Ceramics are also very brittle materials.

Composites

A composite material is a material in which one or more mutually insoluble materials are mixed or bonded together. The primary classes of composites are particulate composites, fibrous composites, and laminated composites.

Mechanical Properties of a Material

The **mechanical properties of a material** are those which affect the mechanical strength and ability of a material to be molded in suitable shape. Some of the typical mechanical properties of a material include:

- Strength
- Toughness
- Hardness
- Hardenability
- Brittleness
- Malleability
- Ductility
- Creep and Slip
- Resilience
- Fatigue

Strength

It is the property of a material which opposes the deformation or breakdown of material in presence of external forces or load. Materials which we finalize for our engineering products, must have suitable mechanical strength to be capable to work under different mechanical forces or loads.

Toughness

It is the ability of a material to absorb the energy and gets plastically deformed without fracturing. Its numerical value is determined by the amount of energy per unit volume. Its unit is Joule/ m^3. Value of toughness of a material can be determined by stress-strain characteristics of a material. For good toughness, materials should have good strength as well as ductility.

Hardness

It is the ability of a material to resist to permanent shape change due to external stress. There are various measure of hardness – Scratch Hardness, Indentation Hardness and Rebound Hardness.

Hardenability

It is the ability of a material to attain the hardness by heat treatment processing. It is determined by the depth up to which the material becomes hard. The SI unit of hardenability is meter (similar to length). Hardenability of material is inversely proportional to the weld-ability of material.

Brittleness

Brittleness of a material indicates that how easily it gets fractured when it is subjected to a force or load. When a brittle material is subjected to a stress it observes very less energy and gets fractures without significant strain. Brittleness is converse to ductility of material. Brittleness of material is temperature dependent. Some metals which are ductile at normal temperature become brittle at low temperature.

Malleability

Malleability is a property of solid materials which indicates that how easily a material gets deformed under compressive stress. Malleability is often categorized by the ability of material to be formed in the form of a thin sheet by hammering or rolling. This mechanical property is an aspect of plasticity of material. Malleability of material is temperature dependent. With rise in temperature, the malleability of material increases.

Ductility

Ductility is a property of a solid material which indicates that how easily a material gets deformed under tensile stress. Ductility is often categorized by the ability of material to get stretched into a wire by pulling or drawing. This mechanical property is also an aspect of plasticity of material and is temperature dependent. With rise in temperature, the ductility of material increases.

Resilience

Resilience is the ability of material to absorb the energy when it is deformed elastically by applying stress and release the energy when stress is removed. Proof resilience is defined as the maximum energy that can be absorbed without permanent deformation. The modulus of resilience is defined as the maximum energy that can be absorbed per unit volume without permanent deformation. It can be determined by integrating the stress-strain cure from zero to elastic limit. Its unit is joule/m^3.

Application of engineering materials

1. Steel
2. Cast iron
3. Copper
4. Aluminum

Mild Steel

- It is a good shock absorber, hence used to make manufacturing screws.
- Universal beams
- Case hardening steel

- Gears

Grey Cast Iron

- Machine tool structure i.e Frame, bed.
- Frames for electrical motors.
- Cylinder blocks and heads for IC engine.
- Lathe machine and drilling machine.

Medium Carbon Steel

- Used to manufacture crankpins, crankshafts
- spline shafts, gear shafts, and axles.
- Also, anchor boats, Axe saw plates, hammers, valve string, and self-taping screws.

High Carbon Steel

- Used in punches and dyes, railway rails, lift leaf springs, and saws for cutting steel and broaches.

Alloy Steel

- Nickel steel is used to manufacture IC engine valves, turbine blades, clock pendulum, measuring instruments.
- Vanadium steel is used to make springs, gears, shafts, and various other tools.

Copper

- Used to make household utensils.
- Used in the manufacturing of electrical cables and wires.
- Used in motor windings.
- Used in condensers and boilers.

Aluminum

Since it has a high resistance to corrosion, it is used to make reflectors and telescopes.

- Used to make foil as food packaging.
- Used in the manufacturing of piston, electric cables, kitchen utensils, automobile parts.

Metal Joining Processes

SOLDERING, BRAZING AND WELDING:

SOLDERING:

Soldering is a method of joining two thin metal pieces using a dissimilar metal or an alloy by the application of heat.

- Temperature is range of 150 to 350 degree.
- Application of flux is externally, usually rosin or borax.
- Soldering application will be electronics circuits.

PRINCIPLE OF SOLDERING

Soldering is very much similar to brazing and its principle is same as that of brazing. The major difference lies with the filler metal, the filler metal used in case of soldering should have the melting temperature lower than 450oC. The surfaces to be soldered must be pre-cleaned so that these are faces of oxides, oils, etc. An appropriate flux must be applied to the faying surfaces and then surfaces are heated. Filler metal called solder is added to the joint, which distributes between the closely fitted surfaces. Strength of soldered joint is much lesser than welded joint and les than a brazed joint.

Advantages of soldering

1) Solder joints are easy to repair
2) Solder joints are corrosion resistance.
3) Low cost and easy to use.
4) Skilled operator is required.

BRAZING:

Brazing is a method of joining two similar or dissimilar metals using a special fusible alloy. The filler metal melts and diffuses over the joint placed.

- The filler metal is called as **Spelters.**
- The flux used is borax or boric acid.

- The brazing is used in copper alloys applications.
- The temperature range is 450 to 900 degree.

PRINCIPLE OF BRAZING

In case of brazing joining of metal pieces is done with the help of filler metal. Filler metal is melted and distributed by capillary action between the faying surfaces of the metallic parts being joined. In this case only filler metal melts. There is no melting of workpiece metal. The filler metal (brazing metal) should have the melting point more than 450^{o} C. Its melting point should be lesser than the melting point of workpiece metal. The metallurgical bonding between work and filler metal and geometric constrictions imposed on the joint by the workpiece metal make the joint stronger than the filler metal out of which the joint has been formed.

BRAZING PROCESSES

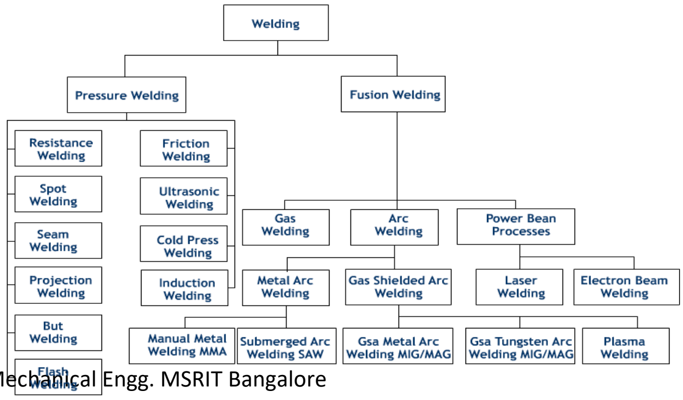
All the processes covered here can also be applied to soldering processes. These common processes are.

1. Torch Brazing
2. Furnace Brazing
3. Induction Brazing
4. Resistance Brazing
5. Dip Brazing
6. Infrared Brazing

WELDING

WELDING: Welding is a process of metallurgically joining two pieces of metals by the application of heat with or without the application of pressure and addition of filler metal. The joint formed is a permanent joint. Modern methods of welding may be classified under two broad headings.

Classification of welding



Types of Welding

1) **Fusion Welding:** joining two metal pieces is heated up to molten state and allowed to solidify, also called as no-pressure welding.

   Ex- Arc welding and Gas welding

2) **Pressure welding:** joining parts to be heated up to plastic state and applying external pressure.

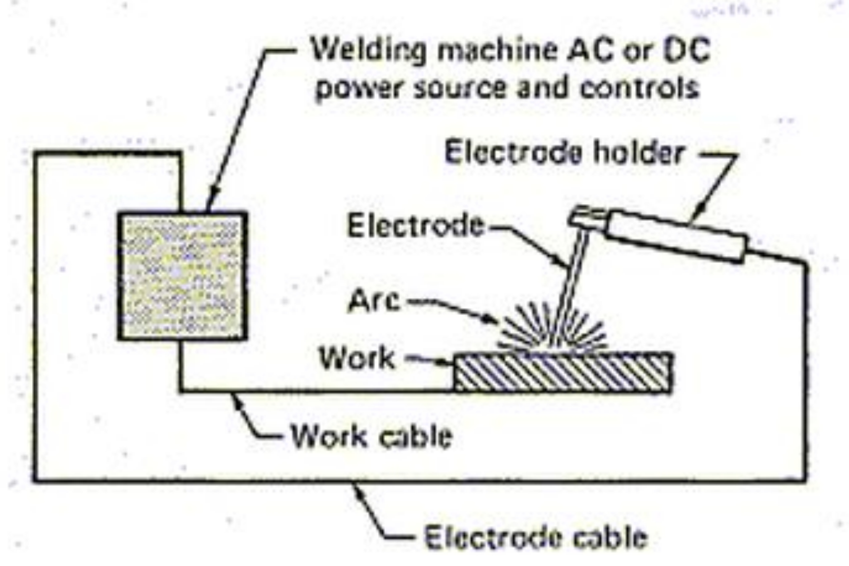
   Ex- Resistance welding and Forge welding.

Electric Arc Welding

The **arc welding equipment** mainly includes an AC machine otherwise DC machine, Electrode, Holder for the electrode, Cables, Connectors for cable, Earthing clamps, Chipping hammer, Helmet, Wire brush, Hand gloves, Safety goggles, sleeves, Aprons, etc.

voltage ranges between 15 – 45 volts,

Range of current is between 30 – 600 amperes.

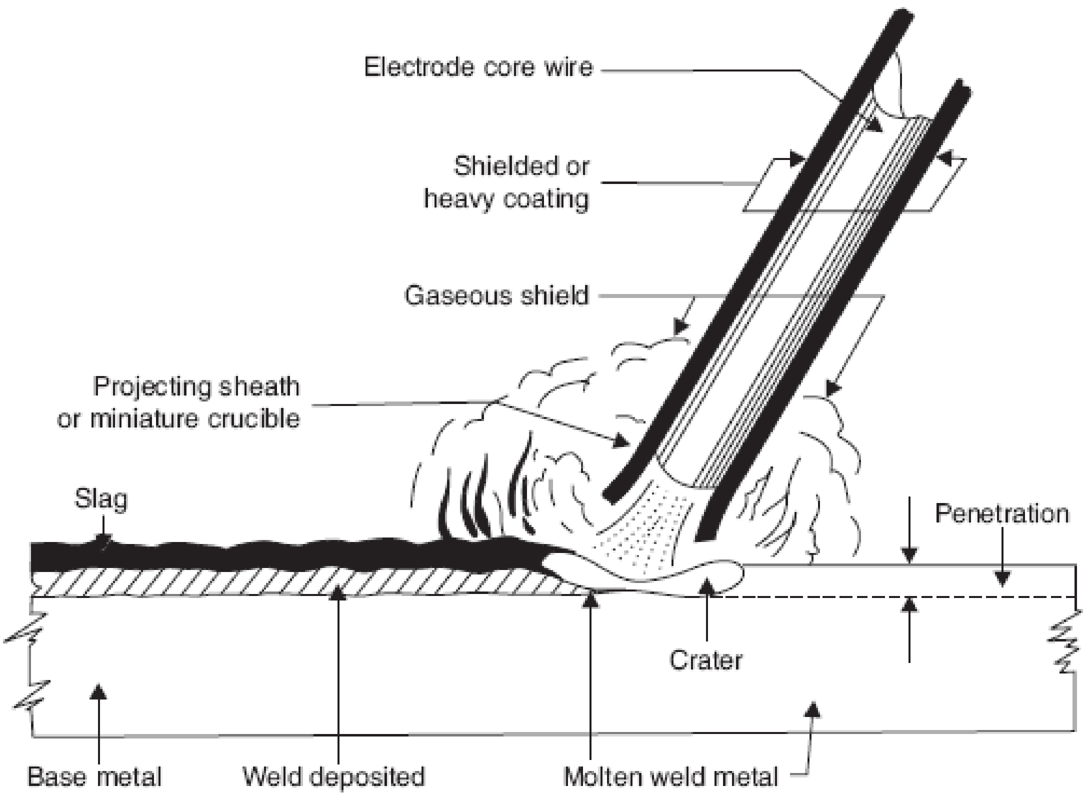


Basic Welding Circuit

The basic arc-welding circuit is illustrated in Fig. 1. An AC or DC power source, fitted with whatever controls may be needed, is connected by a work cable to the workpiece and by a "hot" cable to an electrode holder of some type, which makes an electrical contact with the welding electrode.

An arc is created across the gap when the energized circuit and the electrode tip touches the workpiece and is withdrawn, yet still with in close contact.

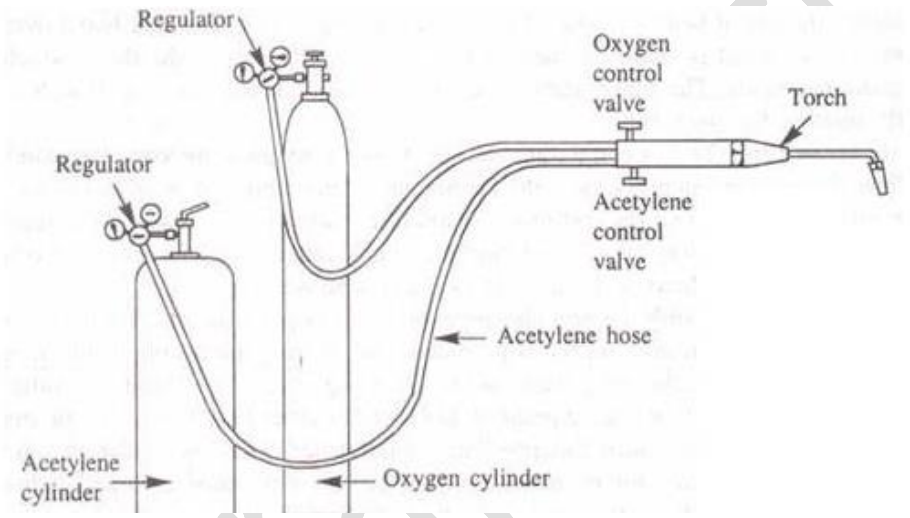
The arc produces a temperature of about 6500ºF at the tip. This heat melts both the base metal and the electrode, producing a pool of molten metal sometimes called a "crater." The crater solidifies behind the electrode as it is moved along the joint. The result is a fusion bond.



A typical arc welding setup is shown in Figure. Arc welding is a method of joining metals with heat produced by an electrical arc. In this process the heat necessary to melt the edges of the metal to be joined is obtained from an electric are struck between the electrode (filler rod) and the work, producing a temperature of 3000-4000^{O} C, in the welding zone. The heat of the arc melts the base metal or edges of the parts fusing them together. Filler metal, usually added melts and mixes with molten base metal to form the weld metal. The weld metal cools and solidifies to form the weld. In most cases, the composition of the filler material, known as welding rod, needed to provide extra metal to the weld, is same as that of the material being welded.

Oxy-Acetylene Welding

Gas welding is a fusion welding process, in which a flame produced by the combustion of gases is employed to melt the metal. The molten metal is allowed to flow together thus forming a solid continuous joint upon cooling. By burning pure oxygen in combination with other gases, in special torches, a flame upto 5000^{O}C can be attained. The gas is purchased in cylinder and connected through resulting valves and pressure gauges into flexible hoses attached to the nozzle. A typical arrangement is shown in Figure



Types of flames:

1. **Neutral Flame**
2. **Carbonizing Flame**
3. **Oxidizing Flame**

1. Neutral Flame

The correct adjustment of the flame is very important for reliable works. When oxygen and acetylene are supplied to the torch in nearly equal volumes, a neutral flame is produced having a maximum temperature of 3200C. This neutral flame is desired for most welding operations. Neutral flame has little effect on the base metal and sound welds are produced when compared to other flames. Figure shows neutral flame.

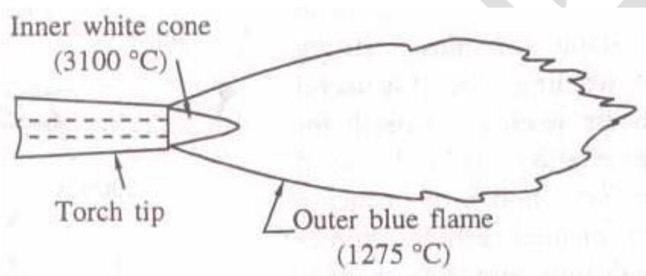


Fig : 1. **Neutral Flame**

2. Carbonizing Flame

In a carbonizing flame or reducing flame excess of acetylene is present. The temperature of this flame is low. The excess unburnt carbon is absorbed in ferrous metals, making the weld hard and brittle. In between the outer blue flame and inner white cone, an intermediate flame feather exists, which is reddish in colour. The length of the flame feather is an indication of the excess acetylene present. Figure shows a carbonizing flame. Carbonizing flame is used for welding high carbon steels and cast iron, alloy steel and for hard facing.

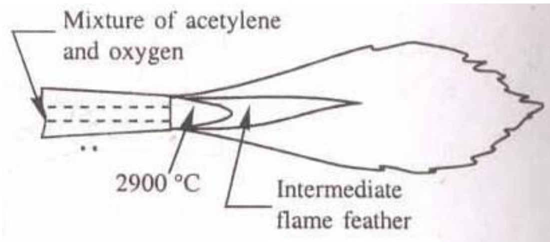


Figure : Carbonizing Flame

3. Oxidizing Flame

In an oxidizing flame excess of oxygen is present. The flame is similar to the neutral flame with the exception that the inner white cone is some what small, giving rise to higher tip temperatures. Excess of oxygen in the oxidizing flame causes the metal to burn or oxidize quickly. Oxidizing flame is useful for welding some nonferrous alloys such as copper and zinc base alloys. The Figure 7 shows the oxidizing flame.

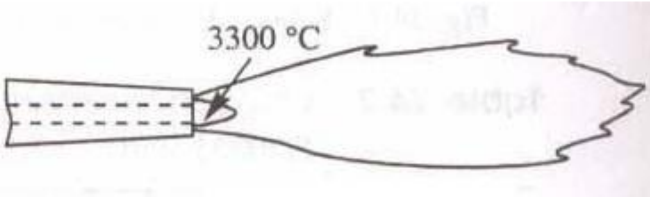


Figure : Oxidizing Flame

Differences between soldering, brazing and Welding

Sl. No.	Welding	Soldering	Brazing
1	These are the strongest joints used to bear the load. Strength of a welded joint may be more than the strength of base metal.	These are weakest joint out of three. Not meant to bear the load. Use to make electrical contacts generally.	These are stronger than soldering but weaker than welding. These can be used to bear the load up to some extent.
2	Temperature required is up to 3800°C of welding zone.	Temperature requirement is up to 450°C.	It may go to 600oC in brazing.
3	Work piece to be joined need to be heated till their melting point.	No need to heat the work pieces.	Work pieces are heated but below their melting point.
4	Mechanical properties of base metal may change at the joint due to heating and cooling.	No change in mechanical properties after joining.	May change in mechanical properties of joint but it is almost negligible.
5	Heat cost is involved and high skill level is required.	Cost involved and skill requirements are very low.	Cost involved and sill required are in between others two.
6	Heat treatment is generally required to eliminate undesirable effects of welding.	No heat treatment is required.	No heat treatment is required after brazing.
7	No preheating of work piece is required before welding as it is carried out at high temperature.	Preheating of work pieces before soldering is good for making good quality joint.	Preheating is desirable to make strong joint as brazing is carried out at relatively low temperature.

RESISTANCE WELDING

The various types of resistance welding are as follows:

(a) Spot welding –

(b) Seam welding

(c) Projection welding

(d) Resistance Butt welding

(e) Flash Butt welding .

(f) Percussion welding.

Principle of Resistance Welding

The name "resistance" welding derives from the fact that the resistance of the workpieces and electrodes are used in combination or contrast to generate the heat at their interface. Heat is generated by the passage of electrical current through a resistance circuit. Heat is generated in localized area which is enough to heat the metal to sufficient temperature so that the parts can be joined with the application of pressure. The force applied before, during and after the current flow forces the heated parts together so that coalescence will occur. Pressure is required throughout the entire welding cycle to assure a continuous electrical circuit through the work. Pressure is applied through electrodes. The pressure is applied by mechanical, hydraulic or pneumatic systems.

1. Resistance Spot Welding (RSW) Working Principle

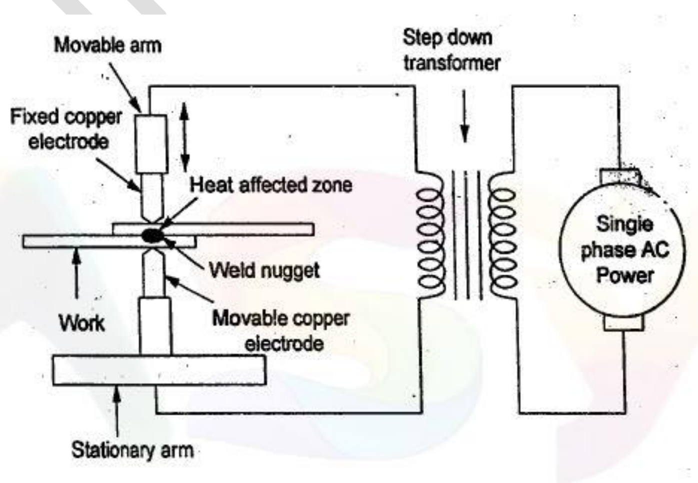
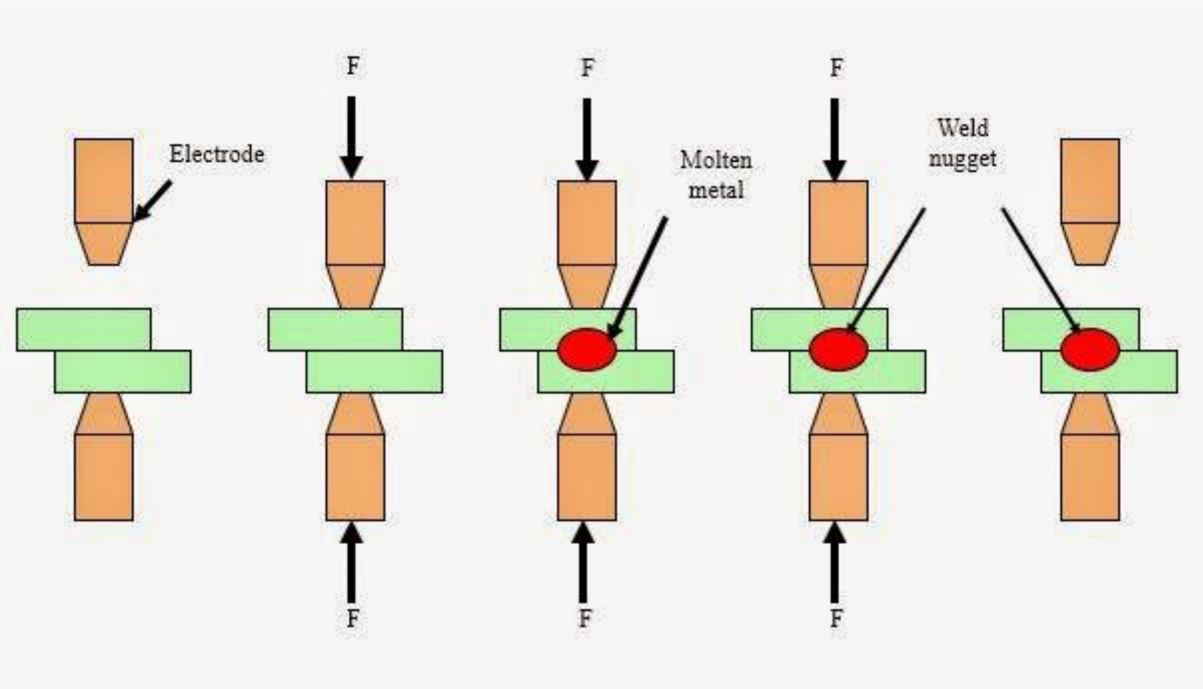


Figure 2.1 Principle of resistance welding



At first the job is cleaned and all types of contaminants like grease, oil, dirt , scale and paint are removed. The surface pf the electrodes are also made very clean. For clamping the metal sheets together two copper electrodes are used at the same time. The current passes through electrodes and then into the metal sheets. Because of the resistance, heat is generated in the air gap within the contact points. Since copper is great conductor heat is dissipated to the metal so quickly. As the metal (workpiece) is a poor conductor of heat in comparison to the copper electrode the heat remains in the air gap. So the heat remains in the one place creating a strong effect and the metal is melted at that desired spot. The period of heat dissipation is very small and at this time metal gets melted and then become solid and thus the joint is formed.

Advantages of Resistance Spot Welding

- Comparatively Low cost
- Resistance Spot Welding (RSW) method doesn't need highly skilled worker.
- Distortion or warping of parts is eliminated though it leaves some depressions or indentation.
- The joint made is highly uniform.
- Automatic or semi-automatic operation both can be done.
- There is no need for edge preparation.
- Welding can be done in quick succession. It just needs a few seconds to make the joint.

Disadvantages of RSW

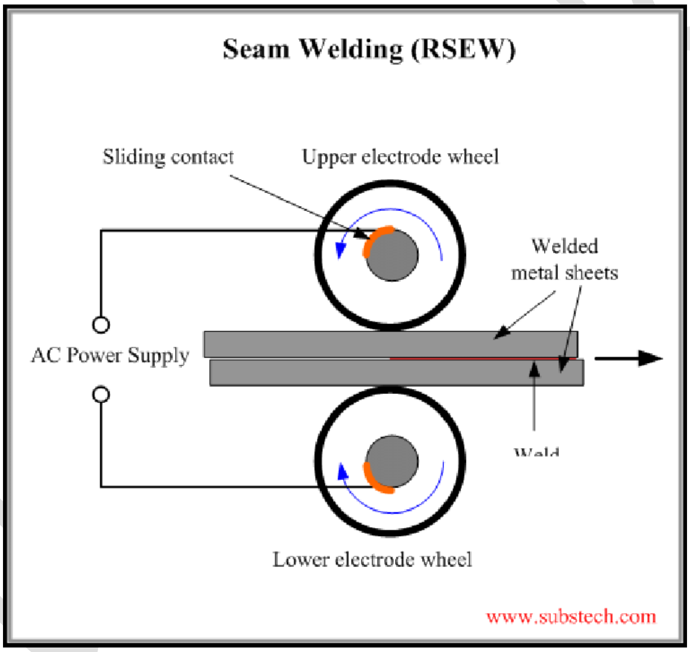
- The equipment cost is high so it can has an effect on the initial cost.
- Skilled welders or technicians are needed for the maintenance and controlling.
- Some metals need special surface preparation for making the RSW a success.

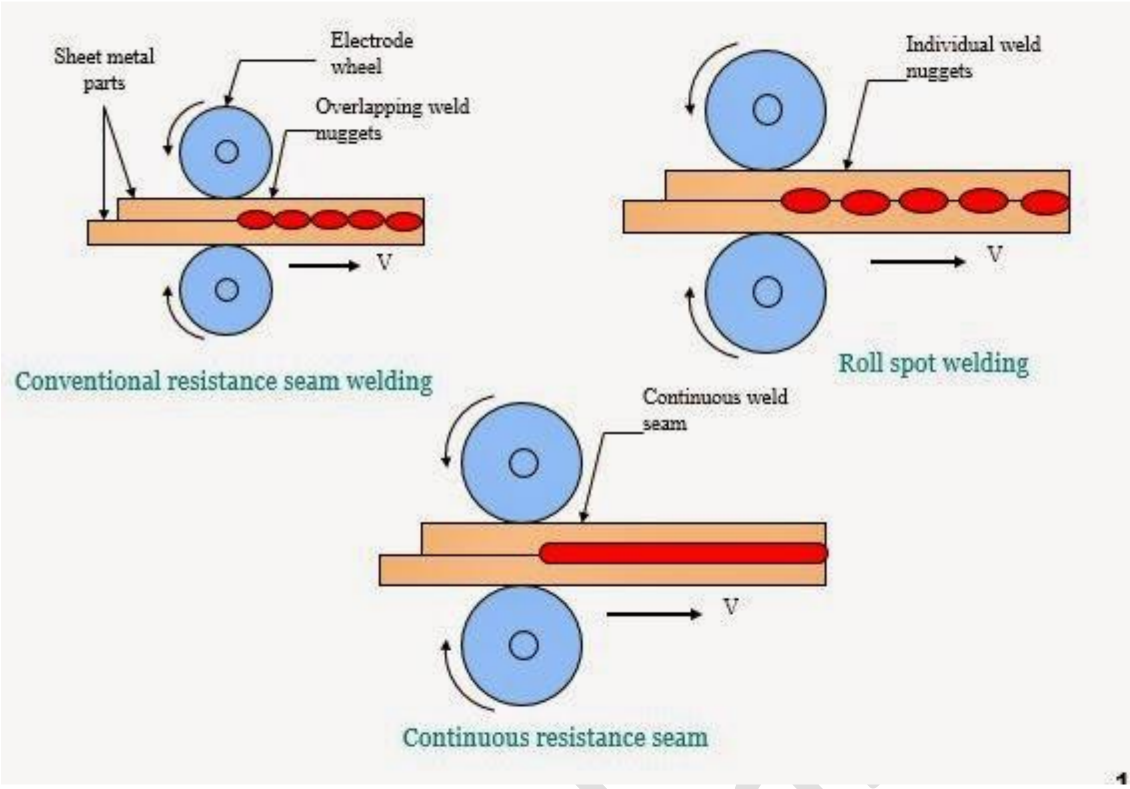
- ➢ The thick jobs are not easy to weld.

Applications of Resistance Spot Welding

- ➢ Spot welding of thick steel plates has been done and it has replaced the need for riveting.
- ➢ The welding of two or more sheet metals can be joined by mechanical means more economically by using the spot welding methods. We don't need gas tight joints.
- ➢ Spot welding can be used for attaching braces, pads or clips with cases, bases and covers which are mainly product of sheet metal forming.
- ➢ Automobile and aircraft industries relies greatly of spot welding these days.

2. Resistance Seam Welding (RSEW) Working Principles





The Resistance Seam Welding (RSEW) is very much similar to the Spot Welding (RSW) but here circular rotating electrodes are used. And here we get continuous weld which is air-tight (If the process is perfect).

The seam-welding form of the resistance process is a series of overlapping welds. Two or more sheets of base metal are usually passed between electrode rollers, as shown in following Figure, which transmit the current and also the mechanical pressure required for producing a welded seam which is normally gas-tight or liquid-tight.

Advantages

- Gas tight as well as liquid tight joints can be made.
- The Overlap is less than spot or projection welding.
- The production of single seam weld and parallel seams can be got simultaneously.

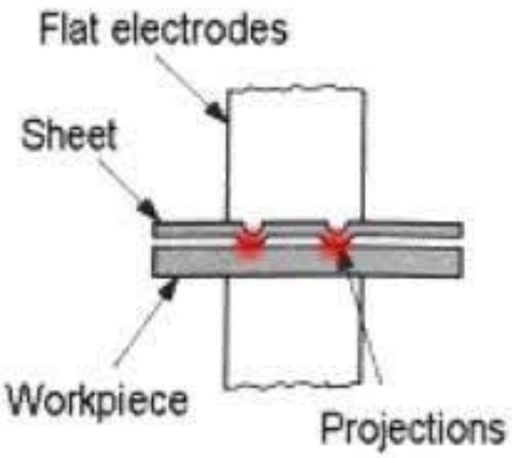
Disadvantages

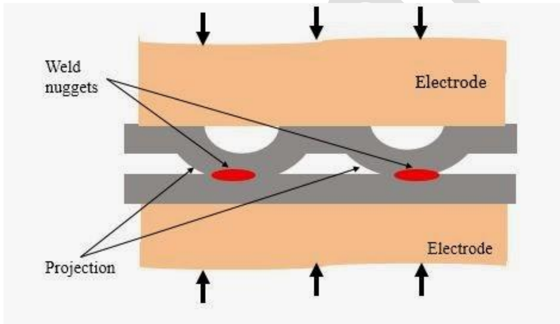
- The welding process is restricted to a straight line or uniformly curved line.
- The metals sheets having thickness more than 3mm can cause problems while welding.
- The design of the electrodes may be needed to change to weld metal sheets having obstructions.

Applications of RSEW

- Girth weld is possible in rectangular or square or even in circular shapes.
- Most of the metals can be welded (Except copper and some high percentage copper alloys)
- Butt welding can be done.

3. Working Principle of Resistance Projection Welding (RPW)





In resistance projection welding (RPW), small projections are formed on one or both pieces of the base metal to obtain contact at a point which localize the current flow and concentrate the heat. Under pressure, the heated and softened projections collapse and a weld is formed. Projection on the upper component is pressed against the lower component by electrode force. The projection collapses and a fused weld nugget are formed with the application of current. This technique is of special value in mounting attachments to surfaces of which the back side is inaccessible to a welding operator.

Advantages of resistance projection welding

- Simultaneous operation can be done i.e. more than one welds can be made.
- Projection welding has this advantage that it can weld metals of thickness which is not suitable for spot welding.
- Projection welding electrodes have a longer life when compared to spot welding electrodes. Its because projection welding electrodes have to withstand less wear and less heating.
- Resistance projection welding is not limited to sheet to sheet joints.
- Projection welding can be done in specific points which are desired to be welded.
- In difficult welding work projection welding gives a better heat balance.
- Projection welding saves electricity because it needs less current to produce heat. So it reduces the shrinkage and distortion defects.

Disadvantages of RPW

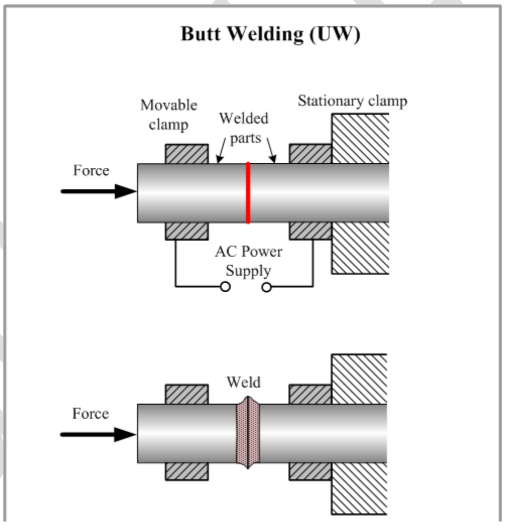
- All types of metals cannot be welded using projection method. Metal thickness and composition is a big question.
- All the metals are not strong enough to support the projections. Some brasses and coppers cannot be welded satisfactorily using projection welding.
- There is an extra operation which is called forming of projection.
- Projections need to have same heights for a appropriate welding.

Applications

- Resistance Projection welding is used in Automobile sector.
- Projection welding is used in refrigeration works (mass production of condensers, gratings, racks etc.)

4. Resistance Butt welding

Resistance Butt Welding is a Resistance Welding (RW) process, in which ends of wires or rods are held under a pressure and heated by an electric current passing through the contact area and producing a weld.

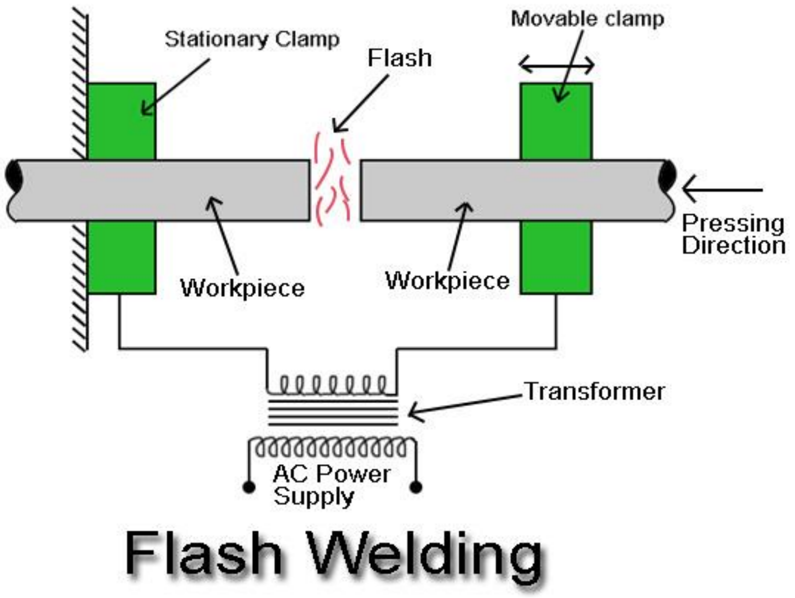


The process is similar to Flash Welding, however in Butt Welding pressure and electric current are applied simultaneously in contrast to Flash Welding where electric current is followed by forging pressure application.

Butt welding is used for welding small parts. The process is highly productive and clean. In contrast to Flash Welding, Butt Welding provides joining with no loss of the welded materials.

5. Flash Butt welding

Flash welding is a method of welding in which the ends of the workpiece are pressed together during the welding process. Flash welding is a type of resistance welding. In this welding process, a heavy current is passed through the joint. Resistance welding is a type of welding in which resistance is used to develop heat. In resistance welding, no filler material is used and no flux is used.



Parts Used in Flash Welding:

1 Transformer:
In flash welding, a transformer is used which is connected to AC supply.

2 AC Supply:
An AC supply is used to provide power to the flash welding or flash butt welding.

3 Clamps:
There are two types of clamp used in Flash Welding:
i) Stationary Clamp
The stationary clamp always remains in a fixed position and always clamps the workpiece.

ii) Movable Clamp
The movable clamp can move about its axis in a reciprocating motion.

Both the clamps are used to hold the workpieces in their position while the welding process is carried out.

Working:

At the starting both parts which are to be joined are placed at some distance and facing each other.

One part is attached to the fixed clamp and another part is fixed to the movable clamp.

Now, the air is present between the two parts and when continuous electricity flows then electric ions start jumping from the part clamped in a fixed clamp towards the part clamped in the movable clamp.

The parts to be joint completes the single turn secondary of a heavy duty transformer.

After that, the part in the movable clamp is slowly brought near to the part clamped in the fixed clamp. As it comes closer and closer, the gas present between the two parts starts getting ionized.

Hence heat is developed in the air present between the two parts to be joined. Now as the heat is developed, then both the ends of the two parts get heated.

Advantages Of Flash Butt Welding:

I) The power required is less in this welding process.

ii) The weld surface is not required to be prepared.

iii) This process is very cheap.

iv) Metals with different melting temperatures can be welded using this flash welding.

v) This process offers strength factor up to 100%.

Disadvantages of flash welding:

i) Chance of fire hazards is high.

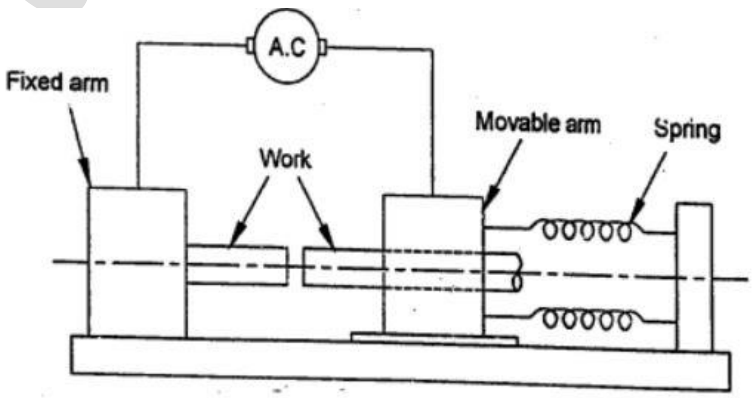
ii) Welding Machine is very bulky.

iii) Metal is lost during flashing and upsetting.

iv) This process is very costly due to high power consumption.

v) The workpiece often loses its straightness during this welding process. The straightness of the workpiece is needed to be maintained.

6. Percussion welding.



It is one type of resistance butt-welding process. The parts to be welded are clamped in copper jaws of the welding machine in which one clamp is fixed and other one is movable. The movable clamp is backed up against the pressure from a heavy spring. The jaws act as electrodes. Heavy electric current is connected to the workpieces. Now, the movable clamp is released rapidly and it moves forward at high velocity. When the two parts are approximately 1.6 mm apart, a sudden discharge of electrical energy is released thereby causing an intense arc between two surfaces. The arc is extinguished by the percussion blow of the two parts coming together with sufficient force to complete in 0.1 second. No upset or flash occurs at the weld. This method is primarily employed to join dissimilar metals. This method is limited to small areas of about 150 to 300 mm2.

Welding energy, E = 1/2 CV2

where
E = energy in watt-seconds (Joules)
C = capacitance in farads
V = voltage.

Amount of energy required to make joint depends on the following factors.
(i) Cross-sectional area of joint
(ii) Properties of work metal or metals
(iii) Depth to which metal is melted on workpieces.

Advantages of Percussion Welding:
1. The time cycle involved is very short.
2. Shortness of arc limits melting and heating.
3. Heat-treated and cold worked materials can be welded without annealing.
4. No filler metal is required.
5. No cast structure is produced at interface.
6.Charging rate is low and controlled.

Limitations of Percussion Welding:

1. The welding process is limited to butt joints.
2.Total area is limited.
3. Similar metals can usually be joined more economically by other processes.
4.The process is usually confined for joining of dissimilar metals not normally considered weldable.
5. Welding is typically dirtier and less smooth than resistance welding.
6. With nib start precussive arc welding, a starting nib must be cut onto workpieces.